

Low Energy Electron Diffraction(LEED)

Advanced Laboratory Course for Master Student,WS 2022

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0.1 Introduction

To investigate the structure of crystalline surfaces we use electrons of lower energy range 50-500 eV. As per the definition of the wavelength, it is inversely proportional to the energy. when we use the electron beam with larger wavelength (i.e. lower energy electron beam), it cannot penetrate deep inside the crystal, but few lattice planes only. So, after the reflection of the electron beam we can get a image of the crystalline surface.

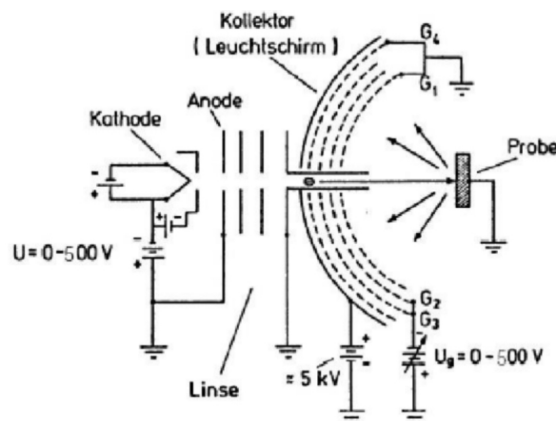


Figure 1: Experimental set up of the LEED optics from LEED script [1]

0.2 Theoretical explanation

After 1924 a French physicist named Louis de broglie proposed a hypothesis about matter waves. which stated that there is a wave associated with a moving particle no matter of what size and defined it through a formula $\lambda = h/p$. Where ($h = 6.626 * 10^{-34} m^2 kg/s$) called Plank's constant. For macroscopic particles the value of wavelength is too small, it can be neglected, but for micro particles like electrons wavelength plays a very crucial role to understand the behavior of these micro-particles.

In this experiment we have to examine the surface structure of copper and the super-structure of attached oxygen using the wave nature of electrons. The electrons of lower energy can penetrate only a few lattice planes of the crystal and get reflected back and gives rise to a diffraction pattern. The energy of electrons is low, so they can only be viewed as non-relativistic particles, their velocity is too small as compered to velocity of light ($v \ll c$). The electron is accelerated with a potential of eV which means the

kinetic energy of an electron is eV. Dealing with the non-relativistic electron, the kinetic energy is written as $p^2/2m$ that follows

$$\lambda = \frac{h}{\sqrt{2meV}} \quad (1)$$

This is elastic scattering, so electrons makes same angle of at reflection as at incidence leading to the Bragg's condition for two parallel electron waves on two parallel lattices. For Braggs condition to follow (constructive interference), an electron should exhibit Laue condition which states that if $\vec{k}_{in}$ is the incident wave vector and $\vec{k}_{out}$ the outgoing wave vector, then the difference is equal to a $\Delta\vec{k}$, is a reciprocal lattice vector equal to $\vec{G}$.

$$\Delta\vec{k} = \vec{k}_{in} - \vec{k}_{out} = \vec{G} \quad (2)$$

Implies that

$$|\vec{G}| = 2|\vec{k}| \sin(\theta) \quad (3)$$

We know $\vec{G} = n \frac{2\pi}{d}$ and $\vec{k} = \frac{2\pi}{\lambda}$, after putting these values in (2) we get the equation exhibiting Bragg's condition. i.e,

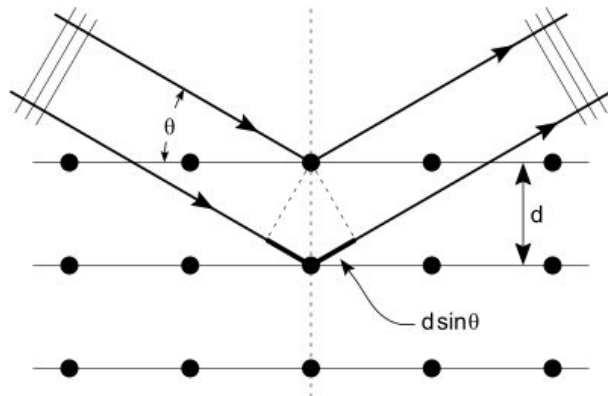


Figure 2: Electron diffraction between two lattice points [2]

$$n\lambda = 2d \sin(\theta) \quad (4)$$

0.3 Experiment Setup

0.3.1 Ultra-High Vacuum Chamber

An ultra-high vacuum chamber is used to perform the LEED experiment in where low pressure less than 10^{-10} (mbar) is maintained to achieve clean surface and almost all the necessary equipment are mounted. This can be done by using following pumping mechanism.

Rotary Vane Pump

The most frequently used pump is rotary vane pump which control gas flow to high pressure to different pressure points. The vane is connected to a rotor, while rotating the system inlet in gas and get compressed on another side. Lastly the compressed gas is put out through the exhaust pipe which is shown in the following figure.

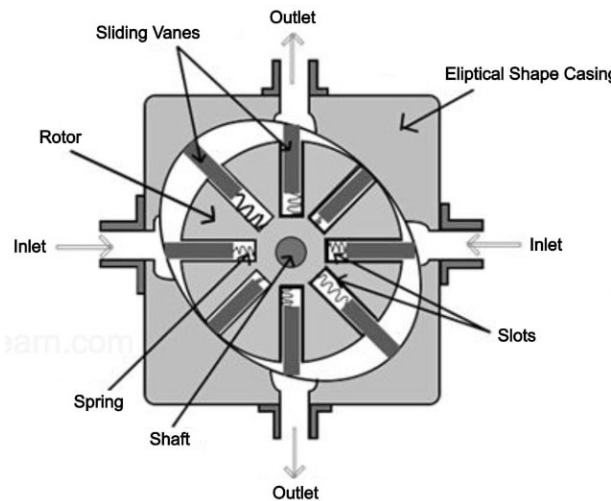


Figure 3: Schematic Diagram of Rotary Vane Pump [3]

Turbo Molecular Pump

Near the main chamber, a Turbo-molecular pump is used to pump the load-lock system to lower the pressure to about 10^{-10} Torr in the Ultra-high vacuum chamber. The Turbo molecular pump consists of motor, rotor, stationary stator blades, inlet and outlet path. As the blades rotates, more and more air get inside and compressed. The movement of the

rotor help to put it into transfer hole and after many cycles,the air exits the chamber through the outlet.

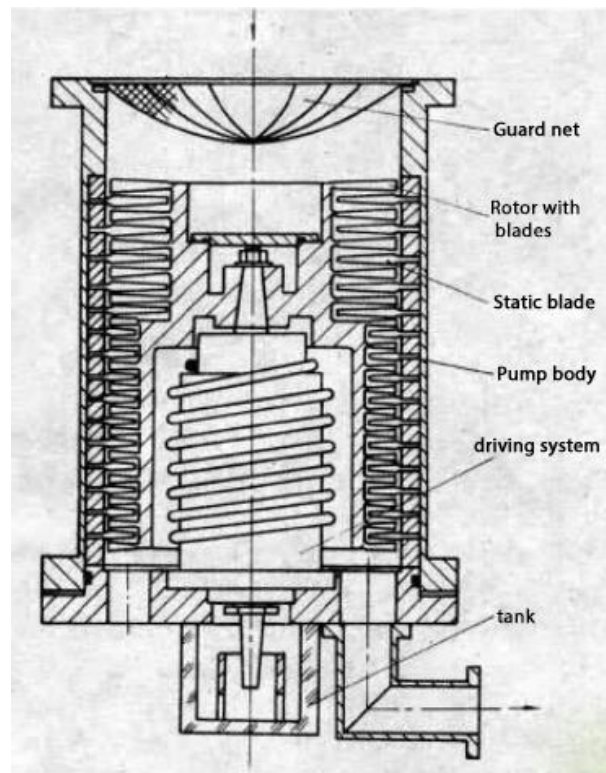


Figure 4: Turbo Molecular Pump [4]

Ion Getter Pump

An ion pump is directly used to evacuate the chamber for fulfilling the empty condition as given below. The Ion pump consists of anode and cathode placed inside a permanent magnet. The inlet air gets ionized while the emitted electrons accelerate from the cathode towards the anode due to intense magnetic field. This ionized atoms then absorbed by the other side of the cathode and eventually removed from the chamber.

- .
- .
- .
- .
- .

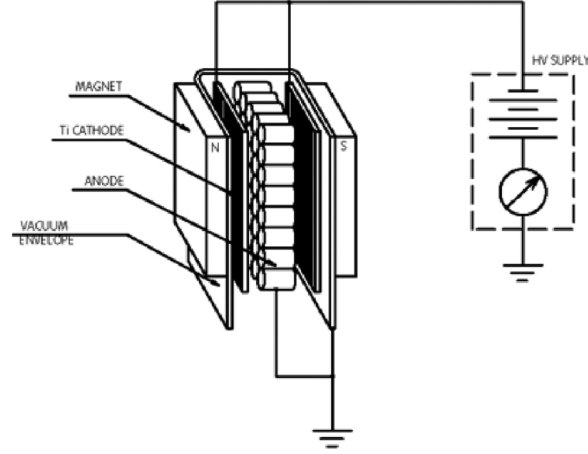


Figure 5: Illustrative graph of Ion getter Pump [5]

Titanium Sublimation Pump

Rest of the remain gas molecules (O_2 , H_2 ..) are removed by using Titanium sublimation pump. A current flow is maintained the to Titanium filament which reaches to it sublimation point and the chamber wall get coated by neat titanium which then react with the residual gas molecules and form stable product. The following figure show the working of titanium sublimation pump.

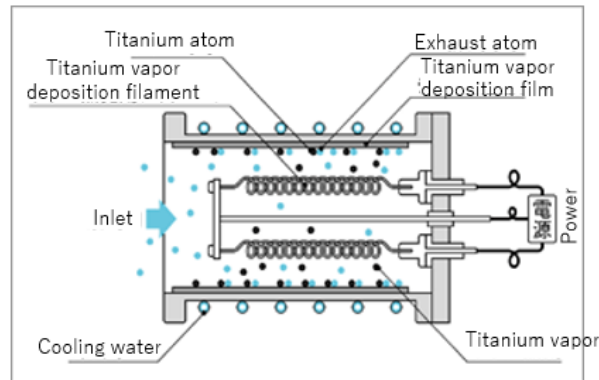


Figure 6: Graphical representation of Titanium Sublimation Pump [6]

0.3.2 Experimental Procedure

Since low pressure is inevitable in LEED experiment to attain as much as a clean surface to avoid interaction of air molecules and particles, the change of pressure is monitored by using a ionization manometer. The sample surface contaminants can be removed

through sputtering device which run the ionized Ar-atoms by using a electrostatic lenses with a field of 2 kV. Now the remain roughness of the sample can be recovered through tempering; a process where the sample heated up(450°C) to a certain temperature for long time. Since the tempering rearrange the sample surface,thus the expected neat and clean sample surface can be obtained. A low energy electron beam, typically in a range of 50-500 eV is used and focused on the sample surface. As the sample used here is a single crystal and well arranged, the incident beam get scattered back and produce diffraction pattern on the screen and recorded by a CCD Camera. The used sample in our experiment is Cu(100) and the size, atomic arrangement and symmetry is measured by recording and analysing various diffracted beams as a function of incident beam energy(I-V curves) [Ref-3]

0.4 Calculations in Advance

For V= 500 V ; K.E = 500 eV

De Broglie Wavelength

$$\begin{aligned}
 \lambda &= \frac{h}{p} \\
 &= \frac{h}{\sqrt{2m_e K.E}} \\
 &= \frac{6.624 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 500 \times 1.6 \times 10^{-19}}} \\
 &= 0.548 \text{ \AA}
 \end{aligned} \tag{5}$$

Relativistic Correction

$$\begin{aligned}
 E^2 &= p^2 c^2 + m_0^2 c^2 \\
 \implies p^2 &= \frac{E^2}{c^2} + m_0^2 c^2 \\
 \implies p &= \sqrt{\frac{(K.E + m_c^2)^2 - m_0^2 c^2}{c^2}} = 120.59 \times 10^{-25}
 \end{aligned} \tag{6}$$

from here we will get the value of wavelength

$$\lambda = \frac{6.624 \times 10^{-34}}{120.59 \times 10^{-25}}$$

$$\implies \lambda = 0.549 \text{ \AA}$$

and it will lead to the equation of speed

$$v = \frac{h}{m\lambda}$$

$$\implies v = 1.32 \times 10^{-7} m/s.$$

The threshold value of Lorentz factor γ to be in non-relativistic limit is 1.1. If this factor exceeds 1.1 then we have to use the relativistic correction. i.e.

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 1.0010 < 1.1 \quad (7)$$

So, there is no need to apply relativistic correction approach, as the speed is too low that will make the Lorentz factor negligible. For an electron to exhibit relativistic character, the energy for its acceleration should be in keV or more than that.

The condition for constructive interference in a scattering event:

It is already calculated in (4) and it states

$$n\lambda = 2d \sin(\theta)$$

Laue condition which states that if $\vec{k}_{in}$ is the incident wave vector and $\vec{k}_{out}$ the outgoing wave vector, then the difference is equal to a $\Delta\vec{k}$, is a reciprocal lattice vector equal to $\vec{G}$.

Validation of Bragg condition for quadratic lattice:

From Bragg's law we already know that

$$n\lambda = 2d \sin(\theta)$$

For 2D we write

$$\vec{G} = n\vec{b}_1 + m\vec{b}_2$$

n	m	V_0
1	1	74.45 eV
0	1	37.22 eV
2	2	297.80 eV

Table 1: Calculation for minimum energy

As

$$d_{n.m} = \frac{2\pi}{|\vec{G}|} = \frac{a}{\sqrt{n^2 + m^2}}$$

$$\implies \sin \theta_{n,m} = \frac{\lambda}{a} \sqrt{n^2 + m^2} \quad (8)$$

Calculation of minimum energy requirement for diffraction:

De Broglie wavelength of the electron beam is

$$\lambda(A) = \frac{12.26}{\sqrt{V(eV)}} \quad (9)$$

Using $a = 2.55$

AA and an opening angle of 52° we can get the following table for the minimum energy V_0

Kinematic Approximation:

The Total energy of an electron can be approximately written as

$$E = E_k - V_0 \quad (10)$$

$$\implies \lambda = \frac{h}{\sqrt{2m_e(E_k - V_0)}} \quad (11)$$

From (4) and (11), we get

$$\implies E_k = \frac{n^2 h^2}{8md^2 \sin^2 \theta} + V_0 \quad (12)$$

0.5 Data Analysis

LEED Images of Cu(100) and lattice constant

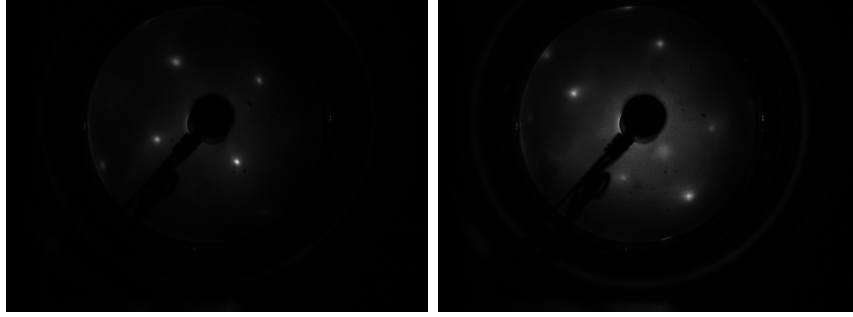


Figure 7: Leed images of Cu(100) surface with energy of 149 eV(left) and 207 eV(right)

The LEED Images can be obtain by varying the incident electron beams energy. In our experiment, we got multiple LEED images in the energy range 50 ev-393 eV and two of them with most bright spots are given below[Fig-8].The spot position is clearly visible from the image at $[-2,0]$, $[0,2]$ and $[2,0]$ for the incident energy 149 eV(left image) and so that gives a LEED pattern $n^2 + m^2 = 4$. Now using equation (8) and (9) we have

$$a = \frac{12.26eV\sqrt{(n^2 + m^2)}}{\sqrt{V(eV)}\sin\theta_{n,m}} \quad (13)$$

By using equation (14), we have lattice constant $a = 3.81 \text{ \AA}$

By closely looking at the LEED image(right) with energy 207eV,the spot position can be determined at $[-1,2]$, $[1,2]$, $[0,2]$ and $[2,-1]$ which turns a pattern of 5. So the lattice constant with the energy 207 eV; we get is $a = 3.53 \text{ \AA}$

Superstructure:

From Woods notation we can derive the superstructure of oxygen on Cu(100)

$$\frac{|\vec{b}_1|}{|\vec{a}_1|} \times \frac{|\vec{b}_2|}{|\vec{a}_2|} R_\alpha \quad (14)$$

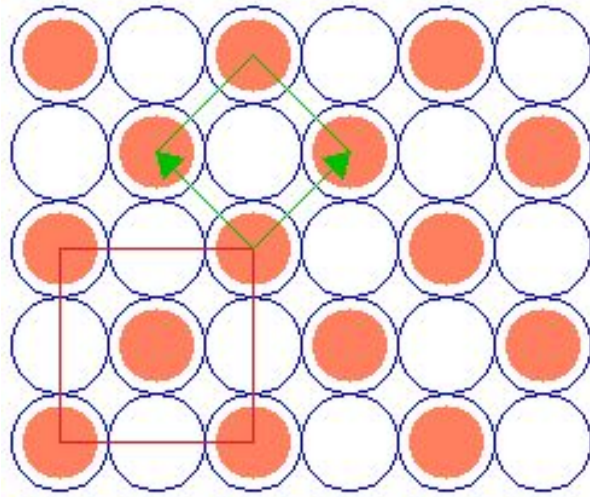


Figure 8: Oxygen Superstructure on Copper Surface with $(\sqrt{2} * 2\sqrt{2})$
R45°[7]

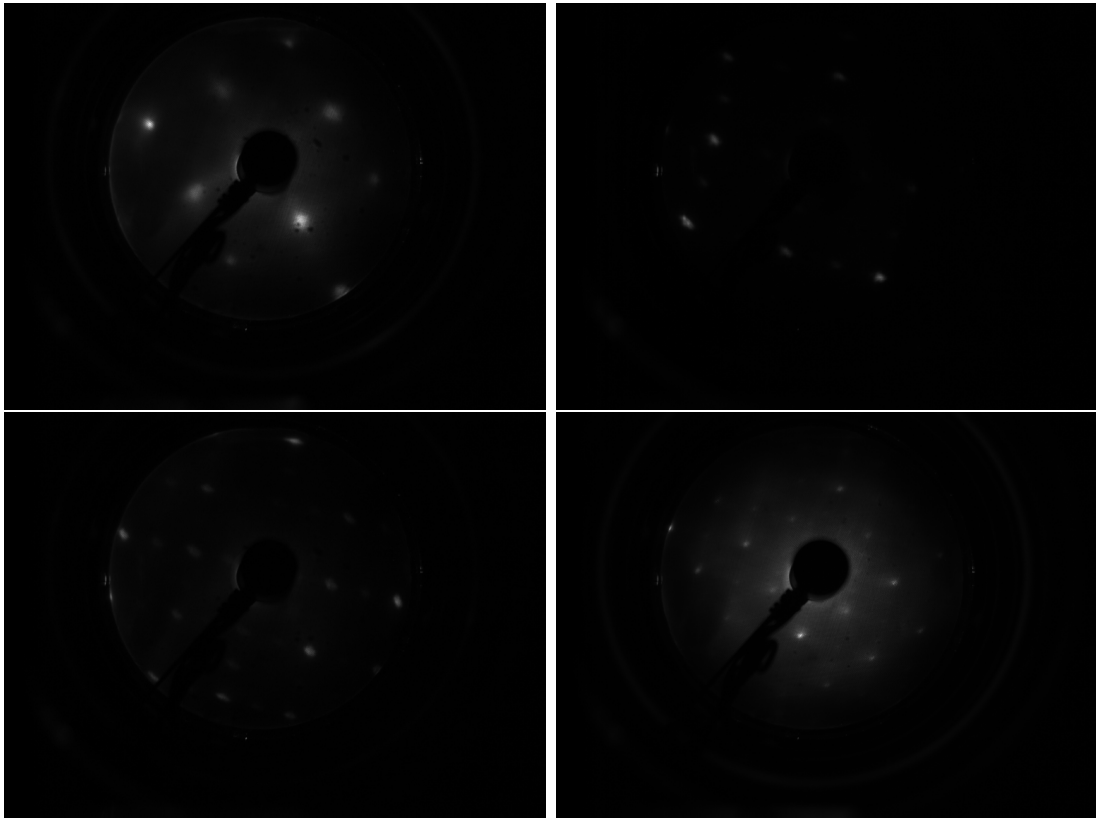


Figure 9: Leed images of Cu(100) surface with energy of 130 eV(top left) and 47 eV(right) as well as 89 eV(bottom-left) and 282 eV(bottom-right)

The superstructures are given in the figure 9. The top right image(47 eV represents diffraction at $[1,0]$, $[0,1]$; the one in the bottom left with energy 89 eV depicts diffraction at $[1, 1]$ and the bottom one in the right show spots at $[2,2]$.From equation equation (14),we can calculate the value of lattice constant 'a' for different values of energies determined in experiment.The values of lattice constant 'a' are given below in a table.

Value of Lattice Constant for Different Energy			
m	n	V_0	a
1	0	47 eV	2.26 Å
0	1	47 eV	2.26 Å
1	1	89 eV	2.34 Å
2	2	282 eV	2.61 Å

I-V Curve Observation

The I-V curve is the prominent way to analysis the surface structure of any solid surface and superstructures.By obtaining the data of various diffracted spots from the lattice relevant to its intensities give precise information about the atomic position.For our case[Cu(100)];the I-V curve contains multiple peaks with background is given in figure (10)

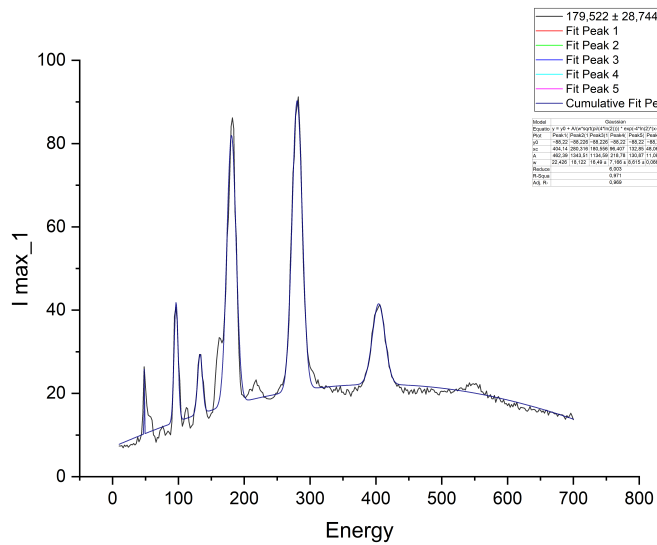


Figure 10: I-V curve for different incident energy[LEED data]

The LEED I-V curves shows various peaks with background or order to multiple scattering events. we are only interested in single scattering at $[0,0]$ position according to kinematic approximation.

$$\Rightarrow E_k = \frac{n^2 h^2}{8md^2 \sin^2 \theta} + V_0 \quad (15)$$

Now the inter-planer distance can be solve by plotting squared of peaks as a function of energy which is shown in figure (11).

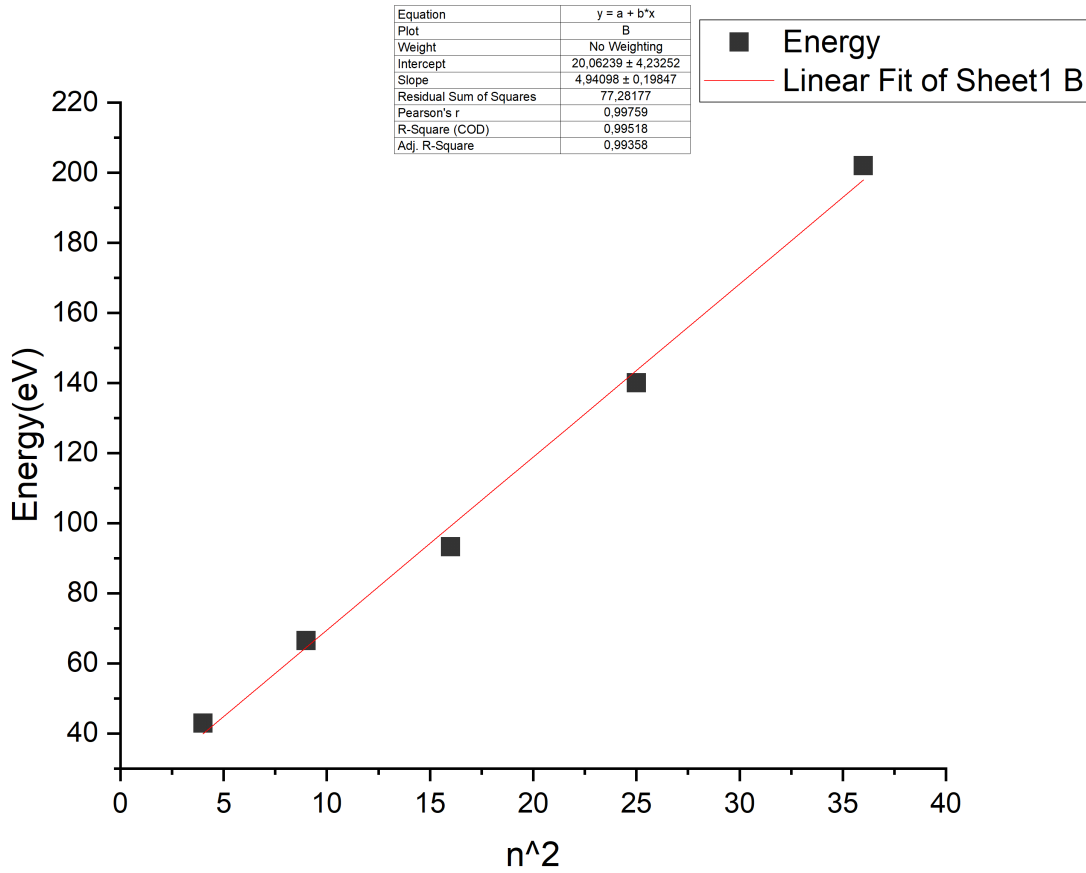


Figure 11: Linear fit of I-V curve $[E \text{ vs } n^2]$

0.6 Conclusion

In this experiment we learned and investigated the structure of Cu(100) using low energy electron diffraction spectroscopy method involving the principle of electron diffraction. For

calculating the values and images of the structure we need a high vacuum chamber and getting that vacuum chamber is too tough, so some necessary measures are taken before and during the LEED experiment to ensure as much as clean chamber and sample surface possible. These measures include annealing by heating the surface of copper upto 800K and to get rid of surface contamination Argon gas is used, this process is termed as sputtering. The experimental results is somewhat deviated from the theoretical value due to the contaminant. The diffraction pattern is somehow vague in the center due to electron gun projection on the screen.

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